TERRAYIELD ANALYTICS PHASE II · ML WORKSTREAM

ML Workshop, Session 1

From canonical data to a benchmark model: turning the Data team's schema into a well-defined ML problem, label strategy, and benchmark.

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Agenda

Eight sections, end to end — from what the Data team hands off to an agreed benchmark specification.

01 Data → ML Handoff

02 ML Problem Definition

03 Canonical ML Schema

04 Feature & Label Strategy

05 Benchmark Targets & Metrics

06 Geospatial Considerations

07 Evaluation Risks

08 Final ML Specification

01 · DATA → ML HANDOFF

What the Data team hands off

Four source schemas, joined on **date + region_id** (crop_type joins price and report data). This is the canonical input before anything is shaped for ML.

Sentinel Satellite

Bands (B02–B08), NDVI/EVI/SAVI, cloud cover, lat/lon, 16-day window

Weather

Max/min temperature, rainfall, humidity, wind speed — daily

Commodity Price

Price per tonne (USD), crop type, market name, reporting cadence

Government Reports

Yield, area harvested, production, season, report type

FROM THE GEOSPATIAL NOTEBOOKS — READING RAW SENTINEL BANDS

```
import rasterio
with rasterio.open("image.tif") as src:
    blue  = src.read(2)   # B02
    green = src.read(3)   # B03
    red   = src.read(4)   # B04
    nir   = src.read(8)   # B08
```

01 · DATA → ML HANDOFF

What we can actually use today

master_dataset.csv is our working reference — not the ML-ready dataset the Data team will eventually deliver. Treat every number below as a note on current state, not a target.

580
rows · 154 columns

2
regions — Iowa, Illinois

2
crops — corn, soybean

72 / 580
rows carry a yield label

Every column from the unified schema is present, but coverage is thin: satellite and weather values are dense, yield and production are sparse, and land-use has no populated column at all yet. Deduplicated to one label per region × crop × season, those 72 rows are **24 independent observations** — enough to define the ML problem and rehearse the pipeline, not enough to train a model we'd trust.

01 · DATA → ML HANDOFF

What's still missing

Three fields the schema needs and doesn't have yet, plus one structural gap.

season_year & forecast_cutoff_date	No first-class time-of-prediction field — we can't yet automatically verify a model isn't looking at the future.
crop_mask_year	Needed once satellite-derived land-cover labels are added, so a model can't use next year's land-cover map to predict this year.
A land-use label column	Doesn't exist in the current dataset in any form — see Section 05.
Crop-level attribution	Corn and soybean rows currently carry identical satellite and weather values for the same region and window — crop_type is a join key without a signal behind it yet.

02 · ML PROBLEM DEFINITION

A handful of task shapes covers most of it

Converting available data into a concrete ML task means picking one of these — domain-agnostic, then applied to our data.

Classification

Location + environmental features → risk category (high/medium/low)

Regression

Geographic + environmental features → a continuous value like yield or demand

Detection

Satellite imagery → find and localize buildings, roads, flooded areas

Segmentation

Satellite imagery → label every pixel with a land-cover class

Forecasting

Past yield + weather trend → next season's value

02 · ML PROBLEM DEFINITION

The two problems we're solving

Regression

Yield Prediction

Inputs — vegetation indices, weather, price and government-report fields, joined on date + region_id (+ crop_type)

Output — yield_tonnes_per_ha (production_tonnes currently excluded, see Section 07)

Classification — proposed

Land Use Detection

Inputs — Sentinel bands / vegetation indices fused with tabular context

Output — land-use / land-cover class — no label source in the dataset yet

03 · CANONICAL ML SCHEMA

What the ML pipeline expects a row to look like

IDs
region_id, crop_type

Spatial
latitude, longitude, window_id / start / end

Timestamps
date, timestamp, report_date, forecast_cutoff_date — proposed

Features
band values, NDVI/EVI/SAVI, weather, price

Target / Label
yield_tonnes_per_ha, production_tonnes, land-use class — proposed

Metadata
schema_version, data_source, extraction_method

03 · CANONICAL ML SCHEMA

One record, two shapes

Illustrative — structure matches the real schema, values simplified for readability.

Canonical record

```
region_id: US-Iowa
crop_type: corn
window_start: 2020-05-08
window_end: 2020-05-24
ndvi_mean: 0.209
evi_mean: 0.209
savi_mean: 0.265
temperature_max_c: 20.6
rainfall_mm: 9.8
price_usd_per_tonne: 168.40
```

ML-ready record

```
region_id: US-Iowa
crop_type: corn
ndvi_mean, evi_mean, savi_mean
temperature_max_c, rainfall_mm
price_usd_per_tonne
season_year — derived, not native
forecast_cutoff_date — not present
target: yield_tonnes_per_ha — present on 72/580 rows
```

Fields in amber don't exist in the current dataset yet — they're what Section 08's open decisions need to resolve.

04 · FEATURE & LABEL STRATEGY

Inputs vs. ground truth

Features — what the model sees

Satellite bands & vegetation indices, weather, price, spatial and time metadata.

Rule: a feature must be observable before the prediction is needed — anything from after that point is a leak, not a feature.

Target — what the model predicts

Yield — yield_tonnes_per_ha, sourced from government agricultural reports.

Land use — no ground-truth source in the dataset yet. Proposed: USDA NASS Cropland Data Layer.

Ground truth means a label that comes from an independent source, not derived from the features themselves — e.g. yield from a government report, not estimated from NDVI.

04 · FEATURE & LABEL STRATEGY

Where labels come from, and where they break

Annotation method

Yield: recorded directly by government surveys — no manual annotation needed. Land use: would require either public raster labels (Cropland Data Layer) or manual polygon annotation over imagery.

Class imbalance

A "did dominant crop change" label was rejected — it rarely flips year to year, so one class dominates before training starts. A continuous cropland share avoids that.

Ambiguous cases

Mixed pixels on county boundaries, cloud-obscured windows, and rotation years all produce records with no single clean label — examples on the next slide.

Conflicting labels

Two label sources can disagree for the same region/season (e.g. state vs. county-level report). Resolution rule and source-of-truth priority: still open — Section 08.

04 · FEATURE & LABEL STRATEGY

Three examples from our own data

NDVI mean falls 0.62 → 0.21 in one window; rainfall 2mm; max temp 34°C	**Read as crop stress, not harvest.** Iowa corn/soybean harvest windows fall later in the season than this one; a rainfall gap this severe with high heat is the stronger explanation.
Dominant crop share in a county: 81% corn → 55% corn, one season to the next	**Read as rotation, not conversion.** A swing this size that recovers the following season fits a normal corn/soybean rotation. Only flag land-use conversion if the share keeps falling in year three.
A 10m pixel straddles a county line, ~50/50 corn and soybean, 42% cloud cover	**No clean label exists.** Forcing a single crop onto a mixed, cloud-affected pixel manufactures a wrong answer that looks right. Correct handling: exclude under a cloud-cover/mixed-pixel threshold, or keep it with a confidence flag — never force a label.

05 · BENCHMARK TARGETS & METRICS

Yield prediction benchmark

Dataset checked against	master_dataset.csv — 580 rows, 154 columns
Primary target	yield_tonnes_per_ha — only target with verified provenance and internally consistent values
production_tonnes	production ≈ yield × area fails across rows where all three are present — consistent with an upstream unit-conversion defect. Excluded as target, feature, and cross-check.
area_harvested_ha	Retained as a dimensional check and candidate area-trend feature, not a target

Label coverage is the binding constraint: the government report block is missing on 508 of 580 rows. The 72 rows carrying yield deduplicate to 24 independent observations — 2 regions × 2 crops × 6 seasons.

05 · BENCHMARK TARGETS & METRICS

Land use benchmark — not yet definable

Blocking issue

No land-use label or usable proxy exists. band_scl is declared as required in the schema but was never materialized — the classification target in the original benchmark goals has no source in the current dataset.

Proposed source

USDA NASS Cropland Data Layer — public domain, annual, continental US, from 2008. 30m resolution through 2023, 10m from 2024; the 30m-resampled series should be adopted across both to avoid a step change in masked pixel counts. Release lag is ~2 months after season end — no layer exists yet for 2026.

Candidate targets, if adopted

crop_area_share
Continuous · primary candidate

crop_area_share_delta
Continuous · change framing

change_magnitude
Continuous · secondary

dominant_transition_class
Categorical · not recommended

06 · GEOSPATIAL CONSIDERATIONS

The spatial properties that change how we evaluate

Coordinates & CRS

Every coordinate needs a reference system. Our data uses WGS84 (EPSG:4326) — lat/lon in degrees, not meters.

Point / Polygon / Raster

Sentinel imagery is raster (a pixel grid); county and field boundaries are vector polygons; weather stations are points.

Spatial features

NDVI or rainfall values are meaningful relative to their location and window — not as standalone numbers.

FROM THE GEOSPATIAL NOTEBOOKS — REPROJECTING BEFORE MEASURING

```
gdf.crs                          # EPSG:4326 — degrees, not meters
gdf_utm = gdf.to_crs(epsg=32615) # reproject to UTM zone 15N (Iowa)
gdf_utm.geometry.area            # now measured in square meters
```

06 · GEOSPATIAL CONSIDERATIONS

Spatial risks in evaluation

Spatial autocorrelation	Nearby locations tend to look alike — similar soil, similar weather — so two "independent" rows from adjacent fields aren't really independent samples.
Spatial leakage	A random train/test split can put two neighboring, near-identical pixels on opposite sides — the model then "predicts" a test point it has effectively already seen next door.
Spatial train/test split	Split by region or county, not by row, so no test location sits next to a training location. This is also why the county-vs-state grain decision matters for evaluation, not just data volume.

07 · EVALUATION RISKS

Six ways a benchmark can look better than it is

Data leakage	A feature's timestamp falls after the row's prediction cutoff — unverifiable today since that cutoff field doesn't exist.
Label leakage	Not identified yet — but no automated rule checks for it either. See Section 08.
Test-set contamination	A row-level split can place near-duplicate satellite windows in both train and test.
Class imbalance	A categorical land-use-change label would be dominated by one class before training starts.
Ambiguous labels	Mixed-pixel, cloud-affected, or boundary-straddling records don't have one correct label.
Spatial leakage	Random splits place neighboring locations on both sides of the train/test boundary.

08 · FINAL ML SPECIFICATION

What's agreed, and what's still open

Agreed

Problem — yield prediction (regression) + land-use detection (classification/share, pending label source)

Inputs — unified schema features, joined on date + region_id (+ crop_type)

Target — yield_tonnes_per_ha (production_tonnes quarantined)

Baseline principle — simple model first, always

Still open

Spatial grain — state or county

Split strategy details

Ingesting the Cropland Data Layer for land-use labels

Primary / secondary metric sign-off

Benchmark performance target

08 · FINAL ML SPECIFICATION

Open questions & next steps

Seven decisions block one or both benchmarks. None of them are modeling questions.

1	Spatial grain — state or county? 24 labeled observations at state grain vs. roughly 4,400 at county grain across ~201 units. Blocks both benchmarks.
2	Ingest the Cropland Data Layer? The land-use benchmark has no target without it.
3	Task type for land use — contingent on the grain decision.
4	Does production_tonnes stay in scope, once its unit-conversion issue is resolved?
5	How do we handle the 2023 → 2024 resolution change in the Cropland Data Layer?
6	How many forecast cutoffs per season, and where?
7	How do we report row count vs. independent label count so results aren't overstated?

Discussion

Where should we start?

Which open decision blocks us the most right now?

Is county-grain recomputation worth the pipeline cost?

Who owns the Cropland Data Layer ingestion decision?